Illuminant Surgical

Skylight: Real-Time Projection Mapping for Surgical Navigation

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[1] ILLUMINANT SURGICAL

Problem

Surgical navigation systems have transformed spinal procedures by enhancing precision and reducing complications through real-time visualization of anatomy relative to tracked instruments [1, 2].

However, widespread adoption remains limited due to invasive tracking methods, high costs, and complex user interfaces [6]. Spine surgery presents high risks— with readmission rates reaching 7.4%, reoperations 16% and complications 18% [3,4,5].

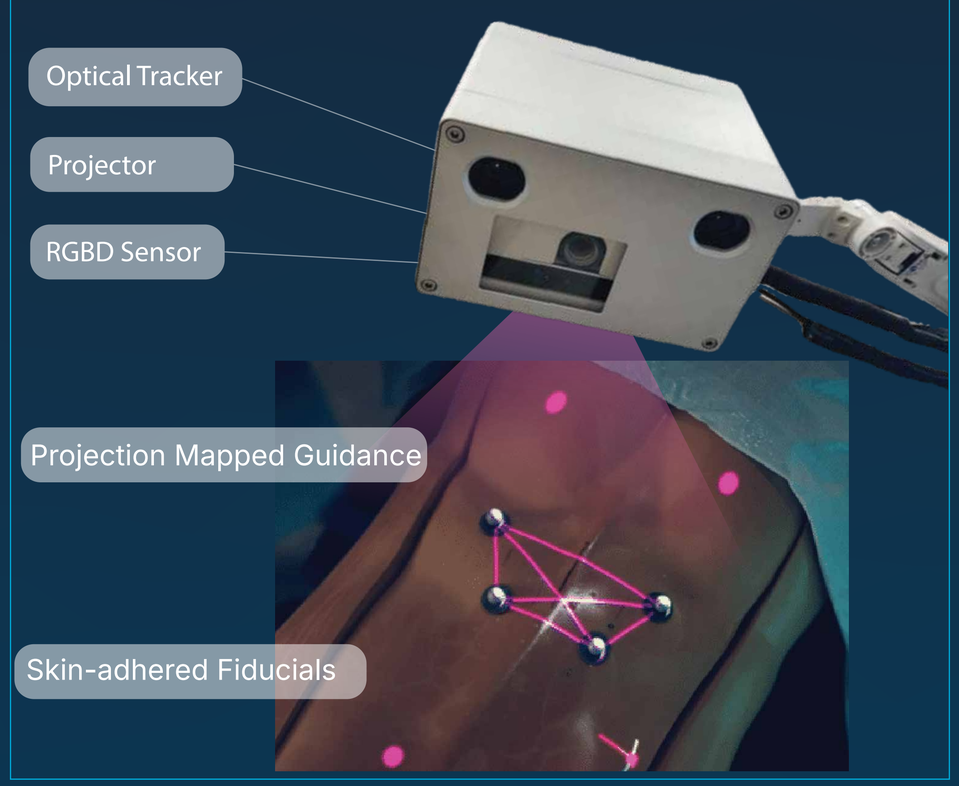
Motivation

Bone-mounted markers are the standard for accurate surgical navigation systems. These are invasive and add signifigant cost and complication, limiting wider adoption.

Visualization relies on displays mounted outside the operating area, or unweildy head-mounted displays. This disrupts focus and provides poor ergonomics.

Our Approach

Skylight adresses limitaitons of existing surgical navigation systems through non-invasive registation and projection-mapped guidance. Skin-adhered fiducials (SkinDots™) are used to register projections directly onto the patient's body in real-time.



References

[1] Suri et al., World Neurosurgery 194.
[2] Bovonratwet et al., Global Spine Journal 13v
[3] Bernatz et al., Neurosurgical Focus 39
[4] Malter et al., Spine 23
[5] Paul, et al. Spine Deformity 4, 4, 304–309.
[6] Choo, et al. International Journal of Spine Surgery

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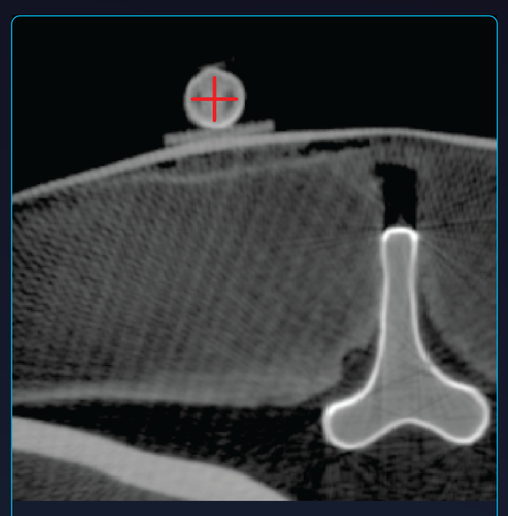
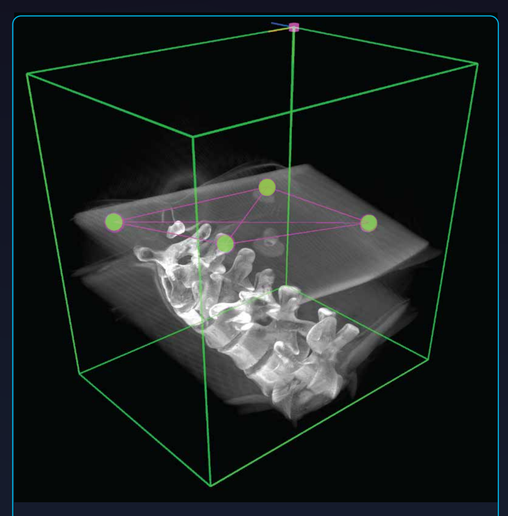
Method

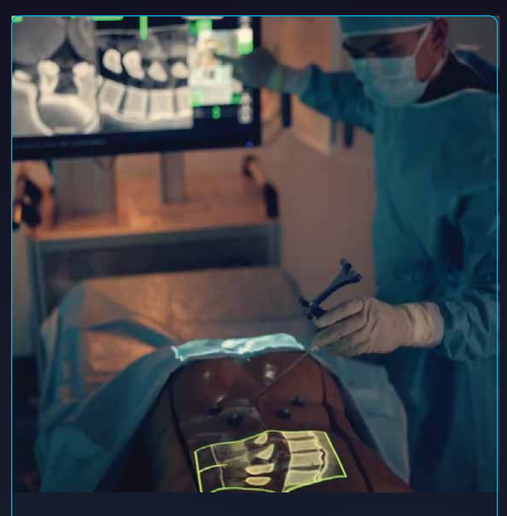
Localize → Register → Track + Project



SkinDots are localized in the CT volume using GPU radial symmetry transform.



The SkinDot pattern is matched to the optical tracker data to register the patient.

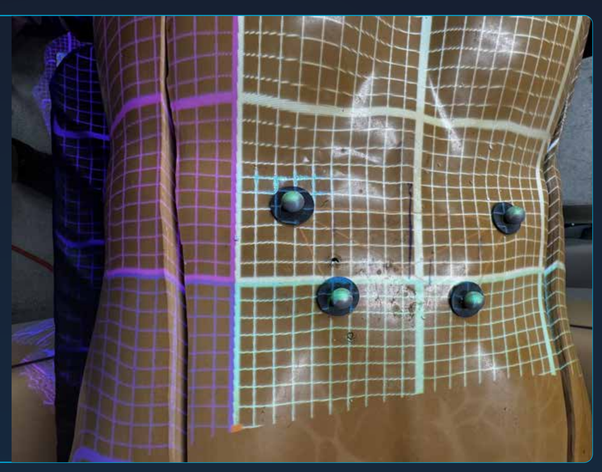


Anatomy and planned trajectories are projected directly onto the live tracked operative area.

Surface Reconstruction

A live 3D reconstruction of the patient's skin is performed using an RGBD sensor. This surface is used to accurately undistort 2D imagery for projection.

Internal anatomy is projected orthographically onto this surface along the CT scan's volume's up axis.



Results

Bench tests on phantom models demonstrate clinical-grade precision, similar to accuracy seen when using bone-mounted fiducials.

Fiducial Localizaiton Accuracy :	Registration Accuracy (RMSE) :	Projection Mapping Error @ 1m:
0.2 mm ± 0.1 mm	**0.3 mm ± 0.15 mm**	**1.5 mm ± 0.7 mm**

Our method provides several key benefits:

- **Headset-free AR:** immersive guidance without bulky wearables.
- **Ergonomically designed**: Maintains natural posture and avoids eye strain.
- **Preserves sterility**: No contact with the surgeon, reducing contamination risks.
- **Line-of-sight visualization:** Projects anatomically aligned data directly onto the patient.